

AIGR Gene Hypothesis Deep Research — Final Report

Target gene: IRE1 (UniProt **G0RBE3**), *Hypocrea jecorina* / *Trichoderma reesei* strain QM6a (NCBITaxon:431241) **Focus type:** function_assignment **Hypothesis slug:** fungal-stress-death-and-signaling-complex **Source file:** `genes/HYPJE/IRE1/IRE1-ai-review.yaml`

Summary

This report evaluates a two-part function-assignment hypothesis for *Trichoderma reesei* / *Hypocrea jecorina* IRE1 (UniProt **G0RBE3**, strain QM6a): (1) that the protein contributes to **intrinsic apoptotic signaling after ER stress** (GO:0070059), and (2) that it forms an **IRE1-TRAF2-ASK1 complex** (GO:1990604). Both propositions were tested independently against direct target evidence, fungal-family evidence, homolog identity, and mechanistic context, exactly as the seed requested. **Both fail.** The verdict is **REFUTED** / **over-annotated by computational carry-over**, with the important nuance that this is a **PAINT/TreeGrafter release-and-source inconsistency (over-propagation from a mammalian subfamily), not a demonstrated experimental loss of an ancestral fungal function.**

The IRE1-TRAF2-ASK1 complex (GO:1990604) is definitionally impossible to assemble in *T. reesei* because two of its three obligate subunits — TRAF2 and ASK1 (MAP3K5) — have no orthologs in the QM6a proteome, and ASK1 is absent kingdom-wide across Fungi (0 hits across taxid 4751). The single MATH/TRAF-domain protein in the proteome (G0RV09) is a ubiquitin carboxyl-terminal hydrolase, not a TRAF signaling adaptor. Every experimental demonstration of the complex in the literature is metazoan (human and rat cell systems). The intrinsic ER-stress apoptotic signaling role (GO:0070059) has never been demonstrated for the fungal protein: the sole primary characterization ([PMID:15480788](#)) establishes an **adaptive UPR** function only — kinase autophosphorylation, complementation of yeast *ire1* mutants, HAC1 mRNA splicing, and chaperone induction — with no assay of IRE1-dependent death.

The two flagged annotations are attached experimentally (IDA) to the **mammalian** PANTHER subfamily **PTHR13954:SF17** (human ERN1/IRE1α, O75460), whereas the fungal target G0RBE3 and the seed's own tree leaf (*Sclerotinia sclerotiorum* A7EHN1) belong to subfamily

PTHR13954:SF6. The fungal proteins therefore carry the terms only as **IEA:TreeGrafter** family-level propagation across a subfamily boundary. The recommended curation action is de-propagation of both apoptosis terms while retaining the evidence-backed core: **Ser/Thr kinase + endoribonuclease** activity driving **IRE1-mediated UPR (GO:0036498)** at the **ER membrane**.

Key Findings

F001 — GO:1990604 (IRE1-TRAF2-ASK1 complex) cannot assemble: two of three obligate subunits are absent from *T. reesei*

The GO term GO:1990604 is defined (QuickGO) as a protein complex composed of IRE1 **plus** TRAF2 **plus** ASK1 — a strict three-subunit definition, so the absence of any single obligate partner is sufficient to refute complex formation. A UniProt REST query of the *Trichoderma reesei* QM6a proteome (taxid 431241) returned **0 proteins** matching TRAF2 / "TNF receptor-associated factor" and **0 proteins** matching ASK1 / "apoptosis signal-regulating kinase" (MAP3K5). The only MATH-domain (IPR002083) protein in the proteome, **G0RV09**, is annotated as a **ubiquitin carboxyl-terminal hydrolase** — it carries the MATH fold but is not a TRAF signaling adaptor and lacks the RING/zinc-finger architecture required for TRAF2 E3/adaptor function.

Every published assay that directly demonstrates the IRE1-TRAF2-ASK1 complex is metazoan: [PMID:23000344](#) (human bladder cancer T24 cells), [PMID:31931206](#) (human SH-SY5Y neuronal cells), [PMID:34427546](#) (rat), and [PMID:33692842](#) (human gastric cancer AGS/MKN-45). As the

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states, "*Ursolic acid induces IRE1-TRAF2-ASK1 signaling complex formation to activate pro-apoptotic ASK1-JNK signaling,*" and

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shows that "*knockdown of IRE1 with the LV-IRE1 shRNA suppressed the expression of IRE1, TRAF2, p-ASK1, and p-JNK*" — both strictly in metazoan cell lines. There is no fungal counterpart.

F002 — GORBE3 is experimentally an adaptive UPR sensor, not a demonstrated apoptosis effector

The single primary characterization of the target protein, [PMID:15480788](#) ("*The ire1 and ptc2 genes involved in the unfolded protein response pathway in the filamentous fungus Trichoderma reesei*"), documents a canonical adaptive UPR sensor and nothing more. It shows: (i) functional complementation of *Saccharomyces cerevisiae* *ire1* mutants; (ii) intrinsic kinase activity — "*The T. reesei IREI protein has intrinsic kinase activity, as revealed by an in vitro autophosphorylation assay*"; (iii) up-regulation of UPR targets *bip1/pdi1* upon *ire1* overexpression; and (iv) HAC1 mRNA splicing — "*Splicing of the mRNA encoding the transcription factor HAC1 is also observed.*"

The UniProt curated record for GORBE3 assigns FUNCTION = ER unfolded-protein sensing + endoribonuclease splicing of precursor mRNAs to induce UPR targets, and SUBCELLULAR LOCATION = ER membrane. The domain architecture (CDD Luminal_IRE1 cd09769, RNase_Ire1 cd10422, kinase PF00069, KEN/PUG) is the canonical UPR sensor. Critically, it contains **no death domain and no dedicated TRAF-recruitment adaptor motif**. No assay of IRE1-dependent apoptotic death has ever been reported in *T. reesei*. The demonstrated outputs (HAC1 splicing, chaperone induction) are the hallmark of the pro-survival, adaptive arm of the UPR, not the terminal/apoptotic arm.

F003 — Fungal programmed cell death runs through metacaspases/mitochondrial factors, not the metazoan IRE1-TRAF2-ASK1-JNK axis

Even granting that fungi can undergo regulated cell death, the executioner machinery differs from the mammalian model. Reviews and primary studies of yeast/fungal PCD attribute apoptosis to the **metacaspases Yca1/Mca1, Nma111**, the DNase **Nuc1**, and the AIF **Ndi1**, acting through **mitochondrial permeability transition** ([PMID:37616576](#); [PMID:22770501](#)). As the direct snippet from

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states, "*Apoptosis is dependent on the Yca1 metacaspase, since loss of YCA1 abrogates cell death induced by oxidized Trx3.*" Because TRAF2 and ASK1 — the obligate partners of the mammalian IRE1 apoptotic complex — have no orthologs in *T. reesei* (0 hits each), the metazoan IRE1→TRAF2→ASK1→JNK death route has no structural basis in this organism. This is a mechanistic, not merely correlative, argument against transferring the mammalian apoptotic role.

F004 — Both flagged terms derive from mammalian primary experiments; ASK1 is absent kingdom-wide in Fungi

The defining primary sources for the IRE1 death module are mammalian. [PMID:10650002](#) (Urano et al., 2000, *Science*) established the IRE1-TRAF2 coupling: "*The cytoplasmic part of IRE1 bound TRAF2, an adaptor protein that couples plasma membrane receptors to JNK activation,*" and showed IRE1 α -/- fibroblasts are impaired in ER-stress JNK activation. [PMID:12050113](#) (Nishitoh et al., 2002, *Genes Dev*) established the complex and its death output: "*ER stress activates ASK 1 through formation of an IRE1-TRAF2-ASK1 complex,*" driving neuronal death from expanded polyglutamine repeats.

Proteome-scale queries confirm the fungal machinery gap. *T. reesei* QM6a encodes only 1 MAP3K-named kinase (**GORDL7, not ASK1**) and 1 MATH-domain protein (G0RV09, a ubiquitin hydrolase, **not TRAF2**). A query for "apoptosis signal-regulating kinase" across the **entire Fungi kingdom (taxid 4751) returned 0 hits** — ASK1/MAP3K5 is a metazoan innovation. Notably, the seed's tree leaf *Sclerotinia sclerotiorum* **A7EHN1** is itself an IRE1 ortholog (InterPro IPR045133; PANTHER PTHR13954:SF6, the same subfamily as G0RBE3), not a TRAF2 or ASK1 protein — confirming the tree neighborhood is composed of fungal UPR sensors, not apoptotic adaptors. This directly answers the seed's question "do target fungal proteins genuinely correspond to TRAF2 and ASK1?" — **No**.

F005 — The experimental complex annotation sits on the mammalian PTHR13954:SF17 subfamily, not the fungal SF6 subfamily of G0RBE3

Human ERN1/IRE1 α (UniProt **O75460**) carries GO:1990604 (IRE1-TRAF2-ASK1 complex) with **IDA** (direct experimental) evidence and GO:0070059 by ISS, and belongs to PANTHER **PTHR13954:SF17**. The fungal IRE1 target G0RBE3 and the seed's tree leaf A7EHN1 both belong to **PTHR13954:SF6**. Thus the experimentally-supported complex annotation resides on a **distinct mammalian subfamily branch**, and the fungal proteins carry the terms only as **IEA:TreeGrafter** family-level (not subfamily-consistent) propagation. This is exactly the release/source inconsistency the seed flags: the current PAINT IBDs for GO:0070059 and GO:1990604 are placed at a mammalian internal node (PTN000359344 per the seed) that is **not an ancestor** of the fungal SF6 leaf.

Programmatic caveat: the PANTHER `family.do` UI returned HTTP 403; the OAI `familymsa` service returned HTTP 200 (2 MB MSA) but full PAINT/IBD internal-node PTN annotations were not machine-parsed in this run. The subfamily assignments above (SF6 vs SF17) are from UniProt/InterPro/PANTHER records rather than a re-derived tree. The exact PTN node IDs (PTN001017826 vs PTN000359344) remain the seed's assertion to verify against the served tree.

Mechanistic Model / Interpretation

The core issue is a mismatch between what the fungal protein *does* and what the mammalian-derived GO terms *require*.

MAMMALIAN IRE1 α (human ERN1, 075460; PTHR13954:SF17)

ER stress $\rightarrow$ IRE1 α oligomerization

(adaptive arm)
XBP1 mRNA splicing
 $\rightarrow$ UPR gene induction

(terminal/apoptotic arm)
cytoplasmic tail binds TRAF2
|
recruits ASK1 (MAP3K5)
= IRE1-TRAF2-ASK1 complex <-- GO:1990604
|
ASK1 $\rightarrow$ JNK $\rightarrow$ apoptosis <-- GO:0070059

FUNGAL IRE1 (T. reesei G0RBE3; PTHR13954:SF6)

ER stress $\rightarrow$ IRE1 oligomerization + autophosphorylation (kinase)

|
HAC1 mRNA splicing (RNase_Ire1 cd10422)

|
Hac1 TF $\rightarrow$ bip1 / pdi1 chaperone induction <-- GO:0036498 (UPR)

|
X NO TRAF2 in proteome (0 hits)
X NO ASK1 in proteome / in ALL Fungi (0 hits, taxid 4751)
X NO death domain / TRAF adaptor motif

Fungal PCD (if any) executes via a SEPARATE route:

oxidative/ER stress $\rightarrow$ mitochondrial permeability transition
 $\rightarrow$ metacaspase Yca1/Mca1, Nma111, Nuc1, Ndi1

The adaptive arm is **conserved** between the mammalian and fungal proteins — both perform kinase autophosphorylation and site-specific mRNA splicing (XBP1 in mammals, HAC1 in fungi) to drive UPR gene induction. The **terminal apoptotic arm is a metazoan elaboration** that depends on partners (TRAF2, ASK1) that fungi never acquired.

Feature	Mammalian IRE1 α (SF17)	Fungal IRE1 (SF6, GORBE3)
Ser/Thr kinase autophosphorylation	Yes	Yes (P 15480788)
Site-specific mRNA splicing	XBP1	HAC1 (P 15480788)
UPR chaperone induction (BiP/PDI)	Yes	Yes (bip1/pdi1)
ER-membrane localization	Yes	Yes (UniProt)
Binds TRAF2	Yes (P 10650002)	No — no TRAF2 in proteome
Recruits ASK1 → IRE1-TRAF2-ASK1 complex	Yes (P 12050113)	No — ASK1 absent kingdom-wide
Drives intrinsic ER-stress apoptosis	Yes	Not demonstrated; machinery absent
PANTHER subfamily	PTHR13954:SF17	PTHR13954:SF6

The correct reading is **conditional/adaptive UPR conservation with loss (or, more accurately, non-acquisition) of the metazoan death-signaling output** — best framed as over-propagation across a family node rather than a demonstrated experimental refutation of a fungal death phenotype.

Evidence Base (Evidence Matrix)

#	Citation	Evidence type	Direction	Claim tested	Key finding
1	UniProt GORBE3 (record)	database/curated	Qualifies	Curated function/ location of target	Ser/Thr kinase/ endoribonuclease IRE1; ER unfolded-protein sensing + HAC1 splicing; ER membrane; apoptosis terms only IEA:TreeGrafter
2	PMID:15480788	direct assay + complementation	Qualifies (supports UPR core; silent on death)	Is IRE1 an adaptive UPR sensor?	Complements yeast <i>ire1</i> ; intrinsic kinase; HAC1 splicing; <i>bip1/pdi1</i> induction
3	UniProt taxid 431241 query (this run)	computational (ortholog absence)	Refutes GO:1990604	Do TRAF2 & ASK1 exist in <i>T. reesei</i> ?	0 TRAF2 hits; 0 ASK1/MAP3K5 hits; sole MATH protein GORV09 is Ub hydrolase
4	QuickGO GO:1990604 definition	database	Refutes	Does complex require TRAF2 + ASK1?	Definition obligately lists IRE1 + TRAF2 + ASK1
5	PMID:10650002	interaction/ mutant	Competing (mammalian origin)	Origin of IRE1→TRAF2→JNK annotation	IRE1 cytoplasmic tail binds TRAF2; IRE1α ^{-/-} impaired in ER-stress JNK
6	PMID:12050113	direct complex/ mutant	Competing (defines GO:1990604)	Origin of complex/death terms	IRE1-TRAF2-ASK1 complex forms; ASK1 essential for ER-stress neuronal death

#	Citation	Evidence type	Direction	Claim tested	Key finding
7	PMID:23000344	direct assay	Competing	IRE1-TRAF2-ASK1 → apoptosis	Complex forms; ASK1-JNK apoptosis
8	PMID:31931206	mutant/ knockdown	Competing	IRE1→TRAF2→ASK1→JNK module	IRE1 knockdown suppresses TRAF2, p-ASK1, p-JNK
9	PMID:33692842	direct assay/co-IP	Competing	IRE1α-TRAF2-ASK1 complex → apoptosis	siIRE1α suppresses complex, blocks apoptosis
10	PMID:34427546	pathway assay	Competing	IRE1-TRAF2-ASK1-JNK activation	Ketamine activates the axis
11	PMID:37616576	review/ phylogenetics	Refutes (fungal death route)	How does fungal apoptosis execute?	Metacaspases Mca1/Nma111, Nuc1, Ndi1 via mito permeability transition
12	PMID:22770501	mutant phenotype	Refutes (fungal death route)	Is yeast apoptosis metacaspase-dependent?	Yca1 loss abrogates oxidative-stress death
13	UniProt taxid 4751 query (this run)	computational (kingdom-wide)	Refutes GO:1990604	Is ASK1 present anywhere in Fungi?	0 "apoptosis signal-regulating kinase" hits kingdom-wide
14	UniProt A7EHN1 (this run)	structural/ evolutionary	Qualifies	Is the tree leaf fungal IRE1?	A7EHN1 is IRE1 (IPR045133; PTHR13954:SF6, same subfamily as GORBE3)

#	Citation	Evidence type	Direction	Claim tested	Key finding
15	UniProt O75460 / human ERN1 (this run)	database (experimental source)	Competing (provenance)	Where is the experimental term?	Human IRE1α GO:1990604 by IDA; PTHR13954: SF17 ≠fungal SF6

How the literature groups. (1) The single fungal-target paper

([P 15480788](#))

supports only the adaptive UPR core and is silent on death. (2) The mammalian mechanistic papers

([P 10650002](#)

[P 12050113](#)

[P 23000344](#)

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[P 33692842](#)

[P 34427546](#))

establish the IRE1-TRAF2-ASK1 apoptotic module exclusively in metazoan cells and are the true origin of the flagged GO terms. (3) The fungal/comparative PCD papers

([P 37616576](#)

[P 22770501](#))

show that fungal regulated cell death executes via metacaspase/mitochondrial factors, providing a positive alternative mechanism that does not require IRE1 nucleation.

GO Curation Implications

Leads requiring curator verification (confirm against the live GO/PANTHER release):

GO ID	Term	Aspect	Current basis	Recommended action	Rationale
GO:1990604	IRE1-TRAF2-ASK1 complex	CC	IEA:TreeGrafter	Remove / de-propagate	TRAF2 + ASK1 absent; ASK1 absent kingdom-wide; complex cannot assemble; IDA basis is mammalian SF17
GO:0070059	Intrinsic apoptotic signaling, ER stress	BP	IEA/ISS carry-over	Remove / mark non-core	No fungal death assay; adaptive UPR only; metazoan-defined process; IBD on mammalian branch
GO:0036498	IRE1-mediated UPR	BP	Experimental (IGI)	Retain	HAC1 splicing + chaperone induction (P 15480788)
GO:0004521 / GO:0004674 / GO:0106310	RNA endonuclease / Ser-Thr kinase	MF	Experimental	Retain (consider more specific endoribonuclease-on-mRNA term)	Autophosphorylation + HAC1 splicing (P 15480788)
GO:0005789	ER membrane	CC	Curated/ISS	Retain	UniProt subcellular location; consistent with domain architecture

Net: the informative retained core is **MF kinase + endoribonuclease** and **BP IRE1-mediated UPR** at the **ER membrane**. This satisfies the objective's requirement to avoid "protein binding" as a terminal recommendation — the retained terms are specific catalytic and process terms.

The two apoptosis terms should be de-propagated, ideally with an annotation-exception flag to block re-propagation, framed as a TreeGrafter over-propagation rather than a demonstrated biological loss-of-function.

Mechanistic Scope

The immediate molecular function under test is whether GORBE3 **directly** (a) participates in a stable IRE1-TRAF2-ASK1 protein complex and (b) initiates an intrinsic apoptotic signaling cascade in response to ER stress.

- **Direct gene-product activity (established):** ER luminal misfolded-protein sensing → oligomerization/auto-activation; trans-autophosphorylation (Ser/Thr kinase); site-specific endoribonucleolytic cleavage of HAC1 mRNA. These are intrinsic, adaptive, homeostatic outputs at the ER membrane.
 - **Complex membership (GO:1990604) — not direct, and physically impossible:** requires partners (TRAF2, ASK1) that are not encoded in the genome. This is not a downstream-phenotype question; it is a hard structural impossibility.
 - **Apoptotic signaling (GO:0070059) — not established:** any conditional fungal death under severe ER stress would, on current evidence, be executed by metacaspase/mitochondrial machinery, not by an IRE1-nucleated TRAF2/ASK1 module. The seed correctly notes that adaptive UPR function and ER localization do not *exclude* conditional death signaling — but the burden of evidence for the specific GO terms is not met, and the required partners are absent.
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Conflicts and Alternatives

1. **Database carry-over (primary alternative explanation).** Both flagged terms are IEA:TreeGrafter. The experimental IRE1-TRAF2-ASK1 evidence belongs to human ERN1 (O75460, PTHR13954:**SF17**); the fungal target is PTHR13954:**SF6**. Family-level propagation crossed a subfamily boundary onto a branch whose ancestor lacks the IBD placement — the best explanation is database carry-over, not biology.

2. **Paralog/component confusion.** The only MATH-domain protein in *T. reesei* is a ubiquitin hydrolase (G0RV09), not a TRAF adaptor; no MAP3K5/ASK1 ortholog exists. Fungal MAP3Ks (Ssk2/Ste11/Bck1) are not ASK1 orthologs and are not ER-stress apoptosis kinases.
3. **Organism-specific machinery differences.** TRAF2 and ASK1 are metazoan innovations; ASK1 returns 0 hits kingdom-wide in Fungi. The mammalian death module has no fungal structural equivalent.
4. **Distinct fungal death route.** If *T. reesei* undergoes regulated cell death, the conserved eukaryotic route is metacaspase + mitochondrial (Yca1/Nma111/Nuc1/Ndi1;

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which does not require IRE1 as a nucleator.

5. **Not a demonstrated loss-of-function.** There is no evidence the fungal lineage ever possessed an IRE1-TRAF2-ASK1 death module, so this is best framed as *absence/never-present in fungi + mispropagation*, not "loss of an ancestral function."

Limitations and Knowledge Gaps

1. **Proteome searches were name/domain-based.** The 0-hit results for TRAF2/ASK1 rely on UniProt annotation text and InterPro/MATH-domain matching, not exhaustive HMM/structural homology profiling. A remote possibility of a highly diverged TRAF-like adaptor missed by keyword search remains. *Resolution:* run HMMER/Foldseek profile searches with TRAF2 (MATH+RING) and ASK1 (MAP3K catalytic + thioredoxin-binding region) profiles against the QM6a proteome and predicted structures.
2. **PANTHER tree not fully re-derived programmatically.** `family.do` returned HTTP 403; the OAI `familysa` service returned HTTP 200 but PAINT/IBD PTN internal-node annotations were not machine-parsed. Subfamily assignments (SF6 vs SF17) are from UniProt/InterPro/PANTHER records. The exact PTN node IDs (PTN001017826 vs PTN000359344) remain the seed's assertion. *Resolution:* obtain the served PTHR13954 v19 tree and PAINT IBD placements directly to confirm the node is not ancestral to the SF6 fungal leaf.
3. **Negative for apoptosis = untested, not disproven.** No *T. reesei* study tested and failed to find IRE1-dependent death. The refutation is definitional (missing partners) plus absence of

positive evidence, not a target-specific negative assay. This distinction matters for how the curation note is worded.

4. **HYPJE-specific PCD literature is thin.** Fungal PCD evidence is largely from *S. cerevisiae* and plants; direct *Trichoderma* PCD-mechanism data are sparse.

Discriminating Tests

Test	What it distinguishes	Predicted result if the seed hypothesis is wrong
HMMER/Foldseek profile search for TRAF2 (MATH+RING) and ASK1 (MAP3K+TBD) vs QM6a proteome + AlphaFold models	Keyword-miss vs true absence of partners	No credible ortholog → complex impossible (supports refutation)
Re-derive served PTHR13954 v19 tree + PAINT IBDs	Whether GO:0070059/1990604 IBD node is ancestral to SF6 fungal leaf	IBD at mammalian node, not ancestral → over-propagation confirmed
<i>T. reesei</i> <i>Δire1</i> survival assay under DTT/tunicamycin ER stress; TUNEL/Annexin-V	Adaptive vs death role of IRE1	<i>Δire1</i> is stress-sensitive (loss of protective UPR), not stress-death-resistant
Co-IP / proximity labeling of GORBE3 under ER stress	Whether any TRAF/ASK-like partner binds fungal IRE1	No TRAF2/ASK1 partner recovered
Metacaspase (<i>yca1</i> -ortholog) epistasis of <i>T. reesei</i> stress death	Whether fungal death uses metacaspase not IRE1-TRAF2-ASK1	Death is metacaspase-dependent, IRE1-independent

Proposed Follow-up Experiments / Actions (Curation Leads)

Candidate action changes (require curator verification) - Remove / de-propagate **GO:1990604** (IRE1-TRAF2-ASK1 complex, CC) from GORBE3; flag as annotation exception to block re-propagation. - Remove or down-grade to flagged non-core **GO:0070059** (intrinsic apoptotic signaling, ER stress, BP) on GORBE3. - Retain and emphasize **GO:0036498** (IRE1-mediated UPR), Ser/Thr kinase + endoribonuclease MF terms, and **ER membrane** CC.

Candidate references with exact snippets to verify -

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— "*The T. reesei IRE1 protein has intrinsic kinase activity, as revealed by an in vitro autophosphorylation assay.*" and "*Splicing of the mRNA encoding the transcription factor HAC1 is also observed.*" → basis for retained UPR core. -

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— "*The cytoplasmic part of IRE1 bound TRAF2...*" → establishes mammalian (not fungal) origin of the TRAF2 link. -

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— "*ER stress activates ASK 1 through formation of an IRE1-TRAF2-ASK1 complex*" → defines GO:1990604 in a mammalian system. -

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— "*Apoptosis is dependent on the Yca1 metacaspase...*" → fungal death route is metacaspase-based.

Suggested curator questions - Is the served PTHR13954 v19 IBD for GO:0070059/GO:1990604 placed at a node ancestral to the fungal SF6 leaf, or only to mammalian SF17? (Expected: only mammalian.) - Should the review record this as a TreeGrafter over-propagation (source inconsistency) rather than a biological loss-of-function claim? (Recommended framing: yes.)

Suggested experiments — see Discriminating Tests table: *Δire1* ER-stress survival/apoptosis assay in *T. reesei*; proteome HMMER/Foldseek sweep for TRAF2/ASK1; served-tree/PAINT re-derivation.

Provenance

- UniProt REST fetch of G0RBE3 (domains, GO codes, curated FUNCTION/CC) — executed this run.
- UniProt proteome queries for TRAF2 / ASK1 / MATH-domain proteins in taxid 431241 — executed this run (0 / 0 / 1 hits; sole MATH hit G0RV09 = Ub hydrolase).
- UniProt kingdom-wide ASK1 query (taxid 4751 → 0 hits) and *T. reesei* MAP3K enumeration (1 hit, G0RDL7) — executed this run.
- A7EHN1 identity check (fungal IRE1, PTHR13954:SF6, IPR045133) and human ERN1 O75460 GO/subfamily check (GO:1990604 IDA; PTHR13954:SF17) — executed this run.
- PANTHER programmatic access test: `family.do` HTTP 403; OAI `familysa` HTTP 200 (PAINT node-level not parsed) — executed this run.
- QuickGO definitions for GO:0070059, GO:1990604, GO:0036498 — executed this run.
- PubMed abstracts reviewed:

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Prepared as a hypothesis-level curation lead. All directional GO recommendations require curator verification against the live GO/PANTHER release. Computational proteome/subfamily results are reported conservatively; the PANTHER tree was not fully re-parsed programmatically in this run (family.do HTTP 403).

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